MILITARY SPECIFICATION

COMPOUNDING OIL, FOR SOLID PROPELLANT GRAINS MARK 75 and MARK 76

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification establishes the minimum requirements for a special grade of low molecular weight napthenic ail of moderate aromatic content to be used as an aid in processing the Mk 75 and Mk 76 solid propellant grains.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government Inspector. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 105), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

ASTM-IP	Petroleum Measurement Standards

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

FSC 1336

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3. REQUIREMENTS

3.1 Preproduction. - The preproduction test sample shall be manufactured using the process methods and equipment proposed for production. Any significant changes after preproduction approval must be approved by the procuring activity prior to incorporation into production. It shall be the responsibility of the contractor to inform the procuring activity of proposed changes after approval of the preproduction sample. In the event of significant changes in the process methods and equipment which in the opinion of the procuring activity may adversely affect the characteristics of the material, additional preproduction samples may be required.

3.2 Chemical and Physical Properties. - Material purchased under this specification shall exhibit the following chemical and physical properties:

TABLE I

CHEMICAL AND PHYSICAL PROPERTIES

Characteristics	Limitations	
	Minimum	Maximum
Specific Gravity at 70°F	.918	.928
Viscosity, SUS at 100°F	153	159
Aniline Point, F	153	159
Refractive Index	1.5036	1.5116

3.3 Workmanship. - The compounding oil shall be a uniform product free from foreign materials. It shall be uniform in quality and manufactured in accordance with standard manufacturing procedures of the industry.

3.4 Data Requirements. - No data is required by this specification or by applicable documents referenced in section 2 unless specified in the contract or order (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of Examinations and Tests. - Examinations and tests of the compounding oil shall be classifed as follows:

(a) Preproduction Tests (4.4)

(b) Quality Conformance Inspection (4.5)

4.3 Lot Size. - A lot shall consist of material produced by one manufacturer in one continuous operation employing not more than one lot of each ingredient and with no changes in formulation or process. If manufacture is by batch process, each batch shall constitute a lot.

4.4 Preproduction Tests. - Preproduction tests shall consist of all of the Quality Conformance Inspections and requirements of this specification.

4.5 Quality Conformance Inspection. - The following procedures shall be performed in duplicate for compliance with section 3.

4.5.1 Sampling. - Sampling for quality conformance inspection shall be in accordance with Standard MIL-STD-105, Inspection Level I, acceptance No. 0, except that the sample size shall never be less than three containers. One pound of material shall be considered as one unit of product (but only for purposes of determining the sample size in applying the above Standard). After the sample size has been determined (based on the total number ot units ot product in the lot), containers equal in number to the sample size shall be selected at random from the lot.

4.6 Specific Gravity.

(a) Pour the sample into a clean 250 ml granduated cylinder or hydrometer jar without splashing so as to avoid the formation of air bubbles. Remove any air bubbles formed, after they have collected on the surface of the sample, by touching them with a piece of clean filter paper before inserting the hydrometer.

(b) Lower a specific gravity hydrometer gently into the sample and, when it has settled, depress it about 2 scale divisions into the liquid and then release. Keep the rest of the stem dry, as unnecessary liquid on the stem changes the effective weight of the instrument, and so affects the reading ob-

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tained. Allow sufficient time for the hydrometer to become completely stationary and for all air bubbles to come to the surface.

(c) When the hydrometer has come to rest, floating freely, and the temperature of the sample is constant to 0.2°F, read the hydrometer to the nearest scale division. To make a reading, observe the point on the hydrometer scale to which the sample rises above its main surface, placing the eye slightly above the plane surface of the liquid.

(d) Observe the temperature of the sample to the nearest 0.25°F immediately before and after the observation of the gravity, the liquid in the cylinder being thoroughly but cautiously stirred with the thermometer, (if the hydrometer has no thermometer, use a thermometer reading degrees Fahrenheit), the whole of the mercury thread being immersed. Should these readings differ by more than 1°F, repeat the temperature and gravity observations when the temperature of the sample has become more stable. Correct the specific gravity reading to 60°F by using Table 23 of the ASTM-IP petroleum measurement tables. Report these results as specific gravity of the sample.

4.7 SUS Viscosity at 100°F. -The following procedure shall be followed to determine SUS viscosity at 100°F:

(a) Make sure the viscometer is perfectly clean before using by washing the viscometer thoroughly with acetone, and letting it air dry. Pour 10 ml of the sample to be tested into the viscometer and immerse the viscometer in the oil bath. Make sure the viscometer is vertical in the bath. Maintain the bath temperature at 100°F ± 0.02°F. Allow 10 minutes for the viscometer and contents to reach equilibrium, and then raise the liquid by means of a suction bulb past the top mark on the viscometer.

(b) Let the liquid fall, and at the same time the meniscus flows past the top mark on the viscometer, start the timer. Continue watching the top meniscus of the liquid, and as it passes the bottom mark on the viscometer, stop the timer. Record the time interval and the temperature of the bath at the end of the test. Repeat the test until there are at least three readings that agree within 0.1 second of each other. Calculate the viscosity, n_L, as follows:

$$n_L = K d_L T_L$$

Where: K = Calibration factor (see calibration procedure) for the viscometer used

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d_L = Density of the liquid (by pyconometer or vendors certification)
T_L = Time for the liquid to flow past the two marks

(c) Calibration Procedure: Repeat the test as above, leaving out the calculations, and using distilled water as the test liquid. Calculate K as follows:

$$K = \frac{NW}{dwTw}$$

Where: NW = Viscosity of water at the temperature of the bath (0.6891 cs)
dw = Density of water at the temperature of the bath (0.9951)
Tw = Time for the water to flow past the two marks

NOTE: The K calculated is for the particular viscometer used, and is accurate onlyfor the temperature at which it was calibrated. Each viscometer has to be calibrated individually.

(d) After the viscosity has been calculated for the Kinematic viscosity, this answer must be converted to SUS (Saybolt Universal Seconds) by using the following equation:

Saybolt Universal Seconds = Centistokes x 4.635

Where: Centistokes = The centistokes found from the Kinematic viscosity tests = n_L d_L

(e) Report this answer as the SUS of the compounding oil.

4.8 Aniline Point.

4.8.1 Apparatus. -The apparatus shown on Figure 1 shall consist of the following:

(a) A test tube approximately 25 mm in diameter and 150 mm in length, made of heat resistanc glass.

(b) Jacket - A jacket approximately 37-42 mm in diameter and

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175 mm in length, made of heat resistant glass.

(c) A manually operated metal stirrer of No. 14 B & S gauge soft-iron wire as shown on the attachment. A concentric ring shall be at the bottom, having a diameter of approximately 19 mm. The length of the stirrer to the right angle bend shall be approximately 200 mm. The right angle bend shallbe approximately 55 mm long. A glass sleeve approximately 65 mm in length and 3 mm inside diameter shall be used as a guide for the stirrer. Any suitable mechanical device for operating the stirrer shall be satisfactory.

4.8.2 Procedure. -The following procedure shall be followed to determine the aniline point of this material:

(a) Clean and dry the apparatus. Pipet 10 ml of aniline (CAUTION: Aniline should not be pipeted directly because of its extreme toxicity. It is also toxic by absorption through the skin, even in very small quantities, and should be handled with great caution) and 10 ml of the dry sample into the air-jacketed tube fitted with stirrer and thermometer. If the material is too viscous for pipeting, weigh to the nearest 0.01 g a quantity of the sample corresponding to 10 ml at room temperature. Center the thermometer in the test tube so that the immersion mark is at the liquid level, making sure that the bulb does not touch the side of the tube.

(b) Stir the sample rapidly, using a 2 inch stroke, avoiding the inclusions of air bubbles, and, if necessary, heating until complete miscibility is obtained. If the aniline sample mixture is completely miscible at room temperature, substitute a nonaqueous cooling bath for the heating bath. Continue stirring and allow the mixture to cool slowly at the rate of 1.0 to 1.8°F per min. Continue cooling to a temperature of 2.0 to 3.5°F below the first appearance of turbidity and record as the aniline point the temperature at which the mixture suddenly becomes cloudy throughout. (The true aniline point is characterized by a turbidity which increases sharply as the temperature is lowered). This will be the minimum equilibrium solution temperature and not the temperature of separation of minor portions of the sample.

(c) Repeat the heating and cooling operations until the range of three successive observations of the aniline point temperature is not greater than 0.2°F. If this range cannot be obtained after five determinations, repeat the test, using fresh quantities of aniline and sample in clean, dry apparatus.

4.9 Refractive Index. -By Abbe' refractometer.

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4.9.1 Preparation. -The following operations shall be performed in preparation for this determination:

(a) Screw in the thermometer provided with the instrument and start water set at 77°F circulating through the prism boxes. Allow 10 to 15 minutes for equilibrium to be obtained.

(b) Be sure that the surfaces of the prisms are perfectly clean by washing the alcohol and carefully wiping them with fibulous paper.

4.9.2 Procedure. -The following procedure shall be followed in determining refractive index.

(a) Place five drops of the liquid to be tested on the ground surface of the lower prism and immediately bring the prisms together and clamp into position. Be sure there is sufficient sample to cover the surfaces of both prisms.

(b) Adjust the mirror below the prisms so that light is directed into them.

(c) Release hand-positioning control clamps and move the lower arm slowly by hand until a dividing line is obtained through the telescope. To be sure that the dividing line is due to refraction, move the mirror slightly; if the dividing line moves, it is not due to refraction. If this is the case, the mirror must be moved enough to clear the dividing line from the field and the true dividing line looked for again.

(d) Adjust the eyepiece tube until the dividing edge of the field and the crosshair of the eyepiece are both in sharp focus.

(e) Adjust the rotating or compensator drum until the dividing line becomes sharp and as free from color as possible.

(f) Tighten the hand-position control clamp and, by means of the line adjustment screw, move the dividing edge until it is in exact coincidence with the center of the crosshair.

(g) The refractive index n_D may now be read directly by observing the scale through the scale magnifier. Take at least 3 readings by approaching from different directions. Record the results, showing the readings to the fourth decimal point.

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4.9.3 Acceptance Criteria. -All test results shall indicate compliance with the requirements of section 3. Failure to meet the requirements shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. -Preservation and packaging shall be level C or as specified (see 6.2).

5.1.1 Level C. -Preservation and packaging shall be in accordance with standard commercial practice to afford protection against damage, contamination of the product by moisture, corrosion, or other foreign matter.

5.2 Packing. -Packing shall be level C or as specified (see 6.2).

5.2.1 Level C. -The product packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. -Unless otherwise specified by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 and shall include but not be limited to the following (see 6.2):

(a) Manufacturer's name and address

(b) Gross and net weights

(c) Lot number and date of manufacture

(d) The number of this specification

(e) Safety precautions or unusual storage requirements

(f) Purchase order or contract number

6. NOTES

6.1 Intended Use. -Material purchased in accordance with this specification is intended for use as an ingredient in solid propellant for rocket

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motors Mk 38 and Mk 39 and such other uses as may be specified by the procuring activity.

6.2 Ordering Data. -Procurement documents shall specify but not be limited to the following information:

(a) Title, number, and date of this specification.

(b) Type and size of container desired.

(c) Place of inspection.

(d) Minimum lot size, if applicable.

(e) Place of delivery.

(f) Request for copies of inspection data, if source inspected.

(g) Selection of applicable levels of packing and packaging (see 5.1 and 5.2)

(h) Marking requirements (see 5.3).

(i) Data requirements.

6.3 Conflicting Requirements. -Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring activity or appointed agent for interpretation and clarification.

6.4 Request for Deviation. -Request for deviation from this specification, applicable drawings, specifications, or materials or processes shall be forwarded to the procuring activity prior to incorporating the desired change into production. All deviations shall be limited to the contract under which they were granted. A deviation is a before-the-fact request.

6.5 Request for Waiver. -Request for waiver from this specification applicable drawings, specifications, or materials shall be forwarded to the procuring activity. A waiver is an after-the-fact request. A waiver request shall include the following:

(a) Exact nature of the defect.

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(b) CD number and defect classification, if any.

(c) Quantity of items involved.

6.6 Value Engineering. -Manufacturers of material covered by this specification are encouraged to submit value engineering suggestions to the procuring activity in an effort to reduce cost.

6.7 Safety Precautions. -The aniline point determination specified in 4.7 shall be performed only by a qualified operator who is thoroughly conversant with the toxic hazards associated with aniline and the safety precautions called for in handling it.

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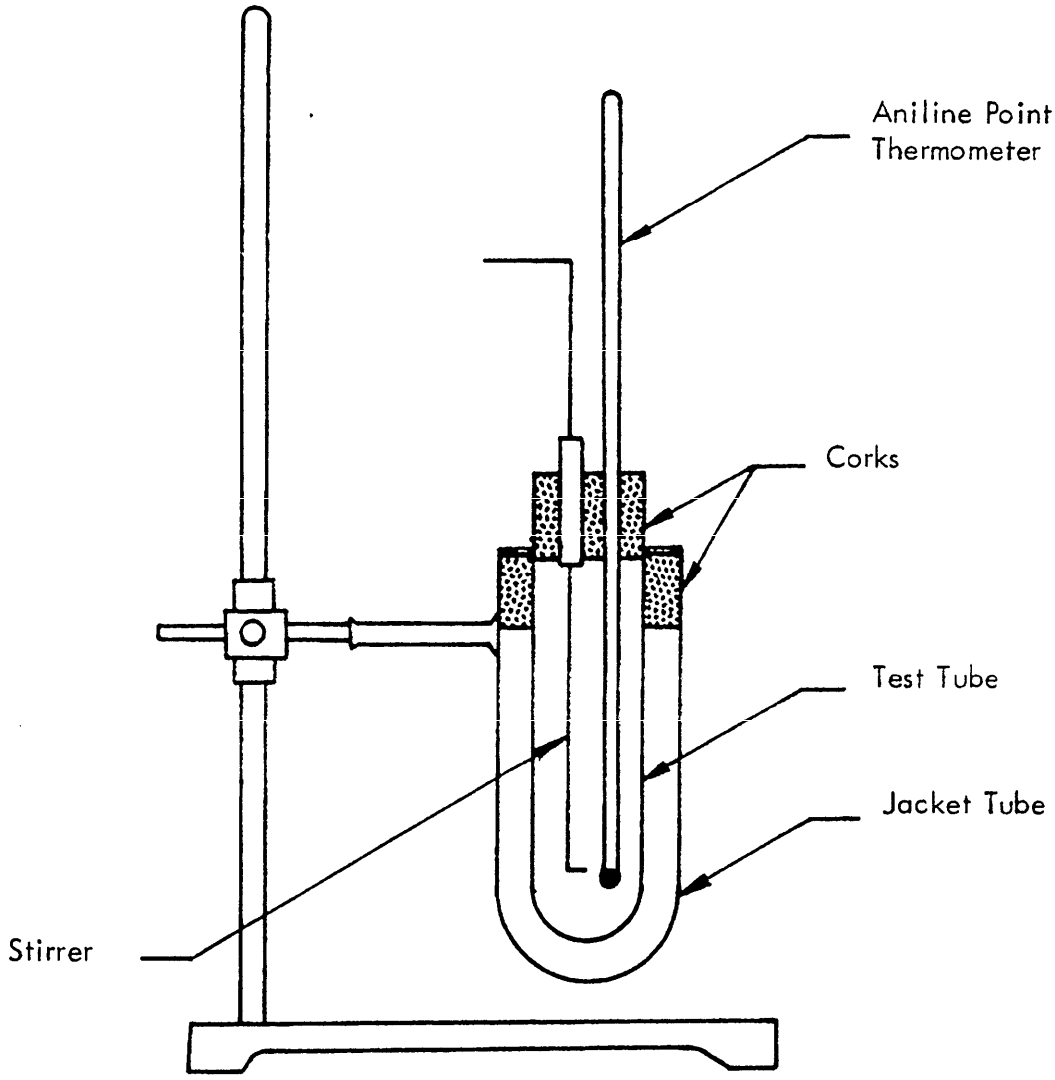


FIGURE 1. ANILINE POINT APPARATUS

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-23947(Wep) COMPOUNDING OIK, FOR SOLID PROPELLANT GRAINS MK 75 & MK 76

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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